

Survey design and analysis using STATA

1. Simple Random Sampling & Weighting

```
*****
* Simple Random Sampling and weight calculation
*****

* Load the dataset, clearing any existing data in memory
use census_sim.dta, clear

* --- Data Preparation and Frame Management ---

capture {
    * Switch back to the default frame
    frame change default
    * Drop the 'census' frame if it exists
    frame drop census
}

frame put *, into(census)
set seed 123

* --- Calculate Population Proportions ---
gen pop = _N

* Calculate the population size for each province
bysort province: gen prov_pop = _N

* Calculate the population size for each municipality
bysort municipality_id: gen munic_pop = _N

* Calculate the proportion of the total population residing in each province
bysort province: gen prov_pop_propotion = prov_pop / pop

* Calculate the proportion of the total population residing in each municipality
bysort municipality_id: gen munic_pop_propotion = munic_pop / pop

* --- Create and Analyze a Sample ---
capture {
    frame change default
    frame drop smpl20
}

frame put *, into(smpl20)
frame change smpl20

sample 20 // Draw a random sample of 20% of the observations from the dataset

gen sample_size = _N

bysort province: gen prov_sample_size = _N
bysort municipality_id: gen munic_sample_size = _N

bysort province: gen prov_sample_propotion = prov_sample_size / sample_size
bysort municipality_id: gen munic_sample_propotion = munic_sample_size / sample_size

* --- Calculate Sampling Weights ---
* Calculate "absolute" or "design" weights for municipalities
* This weight represents how many population units each sample unit represents
gen abs_weight_munic = munic_pop / munic_sample_size

* Calculate "proportional" or "balancing" weights for municipalities
```

* This weight adjusts the sample proportions to match the population proportions
`gen prop_weight_munic = munic_pop_propotion / munic_sample_propotion`

* --- Survey Data Analysis ---

`egen total_pop = total(abs_weight_munic)`

`svyset [pweight = abs_weight_munic]`

`frame census: mean income`

`mean income`

`svy: mean income`

`. frame census: mean income`

Mean estimation

Number of obs = 1,000,000

	Mean	Std. err.	[95% conf. interval]	
income	391705.3	269.7105	391176.7	392233.9

`. mean income`

Mean estimation

Number of obs = 200,000

	Mean	Std. err.	[95% conf. interval]	
income	392632.9	604.0539	391449	393816.8

`. svy: mean income`

(running mean on estimation sample)

Survey: Mean estimation

Number of strata = 1

Number of obs = 200,000

Number of PSUs = 200,000

Population size = 1,000,000

Design df = 199,999

	Mean	Linearized std. err.	[95% conf. interval]	
income	392505.2	604.8006	391319.8	393690.6

`frame census: proportion urban`

`proportion urban`

`svy: proportion urban`

* --- Alternative Weighting Analysis ---

`frame census: sum income`

`sum income`

`sum income [aweight = prop_weight_munic]`

```
+-----+
| [Simple Random Sampling & Weighting] |
+-----+
```

|

```

V
+-----+
| 1. Initial Setup      |
+-----+
|
|   +-- Load Data (`use census_sim.dta, clear`)
|   V
+-----+
| 2. Frame Management   |
+-----+
|
|   +-- Cleanup Old Frames (`capture frame drop`)
|       (Attempts for 'census' & 'smpl20')
|   +-- Create Full Data Copy (`frame put *, into(census)`)
|   +-- Set Random Seed (`set seed 123`)
|   V
+-----+
| 3. Population Calculations (in Default/Original) |
+-----+
|
|   +-- Total Pop Size (`gen pop = _N`)
|   +-- Per Province:
|       |
|       +-- Pop Size (`bysort province: gen prov_pop = _N`)
|       +-- Pop Proportion (`bysort province: gen prov_pop_propotion = ...`)
|   +-- Per Municipality:
|       |
|       +-- Pop Size (`bysort municipality_id: gen munic_pop = _N`)
|       +-- Pop Proportion (`bysort municipality_id: gen munic_pop_propotion = ...`)
|       V
+-----+
| 4. Sample Creation & Analysis |
+-----+
|
|   +-- Create Sample Frame Base (`frame put *, into(smpl20)`)
|       (Copies data *with* population calculations)
|   +-- Switch to Sample Frame (`frame change smpl20`)
|   +-- Draw 20% Sample (`sample 20`)
|       (Reduces observations *within* 'smpl20')
|   +-- Calculate Sample Sizes (in 'smpl20'):
|       |
|       +-- Total Sample Size (`gen sample_size = _N`)
|       +-- Per Province (`bysort province: gen prov_sample_size = _N`)
|       +-- Per Municipality (`bysort municipality_id: gen munic_sample_size = _N`)
|   +-- Calculate Sample Proportions (in 'smpl20'):
|       |
|       +-- Per Province (`bysort province: gen prov_sample_propotion = ...`)
|       +-- Per Municipality (`bysort municipality_id: gen munic_sample_propotion = ...`)
|       V
+-----+
| 5. Calculate Sampling Weights (in 'smpl20') |
+-----+
|
|   +-- Absolute/Design Weight (Municipality)
|       | (`gen abs_weight_munic = munic_pop / munic_sample_size`)
|       | (Represents # of pop units per sample unit)
|   +-- Proportional/Balancing Weight (Municipality)
|       | (`gen prop_weight_munic = munic_pop_propotion / munic_sample_propotion`)
|       | (Adjusts sample distribution to match population)
|       V
+-----+
| 6. Survey Data Analysis |
+-----+

```

```

|
+-- Estimate Total Pop from Sample
|   (`egen total_pop = total(abs_weight_munic)`)
|
+-- Declare Survey Design (in 'smpl20')
|   (`svyset [pweight = abs_weight_munic]`)
|
+-- Compare Population vs. Raw Sample vs. Weighted Sample:
|   +-- Mean Income (`frame census: mean income` vs `mean income` vs `svy: mean income`)
|   +-- Proportion Urban (`frame census: proportion urban` vs `proportion urban` vs `svy:
proportion urban`)
|       V
+-----+
| 7. Alternative Weighting Analysis |
+-----+
|
+-- Compare Total Income:
|   +-- Population (`frame census: sum income`)
|   +-- Raw Sample (`sum income` in 'smpl20')
|   +-- Prop. Weighted Sample (`sum income [aweight=prop_weight_munic]` in 'smpl20')

```

2. Stratified Sampling (Fixed Count per Municipality)

```

*****
* Drawing equal samples from each municipality
*****

frame change census
capture {
    * Switch back to the default frame
    frame change default
    * Drop the 'smpl20' frame if it exists
    frame drop smpl_fixed
}
frame put *, into(smpl_fixed)

frame change smpl_fixed
set seed 123
bysort municipality_id: sample 100, count

gen sample_size = _N

bysort province: gen prov_sample_size = _N
bysort municipality_id: gen munic_sample_size = _N

bysort province: gen prov_sample_propotion = prov_sample_size / sample_size
bysort municipality_id: gen munic_sample_propotion = munic_sample_size / sample_size

gen abs_weight_munic = munic_pop / munic_sample_size
gen prop_weight_munic = munic_pop_propotion / munic_sample_propotion

* --- Survey Data Analysis ---
egen total_pop = total(abs_weight_munic)

svyset [pweight = abs_weight_munic]
frame census: mean income
mean income
svy: mean income

```

```
. frame census: mean income
```

```
Mean estimation                               Number of obs = 1,000,000
```

	Mean	Std. err.	[95% conf. interval]	
income	391705.3	269.7105	391176.7	392233.9

```
. mean income
```

```
Mean estimation                               Number of obs = 75,300
```

	Mean	Std. err.	[95% conf. interval]	
income	356734	926.8036	354917.4	358550.5

```
. svy: mean income
```

```
(running mean on estimation sample)
```

```
Survey: Mean estimation
```

```
Number of strata =      1          Number of obs   =    75,300
Number of PSUs   = 75,300          Population size = 1,000,000
                                   Design df        =    75,299
```

	Mean	Linearized std. err.	[95% conf. interval]	
income	392184.5	1085.448	390057.1	394312

```
frame census: proportion urban
```

```
proportion urban
```

```
svy: proportion urban
```

```
* --- Alternative Weighting Analysis ---
```

```
frame census: sum income
```

```
sum income
```

```
sum income [aweight = prop_weight_munic]
```

```
+-----+
| [Stata Script: Stratified Sampling (Fixed Count per Municipality)] |
+-----+
|
| V
+-----+
| 1. Frame Management (Start from 'census' frame) |
+-----+
|
| +-- Cleanup Old Frame (`capture frame drop smpl_fixed`)
| | (In 'default' frame temporarily)
| |
| +-- Create New Frame Base (`frame put *, into(smpl_fixed)`)
| | (Copies 'census' data, including pop variables)
| |
| +-- Switch to New Frame (`frame change smpl_fixed`)
| |
| +-- Set Random Seed (`set seed 123`)
|
```

```

|
V
+-----+
| 2. Sampling (Fixed Count per Stratum/Municipality) |
+-----+
|
|  +-- Draw Fixed Sample per Municipality (`bysort municipality_id: sample 100, count`)
|  | (Takes 100 obs from each municipality, if available)
|  |
|  V
+-----+
| 3. Sample Analysis (in 'smpL_fixed' frame) |
+-----+
|
|  +-- Calculate Sample Sizes:
|  |
|  |  +-- Total Sample Size (`gen sample_size = _N`)
|  |  +-- Per Province (`bysort province: gen prov_sample_size = _N`)
|  |  +-- Per Municipality (`bysort municipality_id: gen munic_sample_size = _N`)
|  |  | (Note: munic_sample_size will be <= 100)
|  |
|  +-- Calculate Sample Proportions:
|  |
|  |  +-- Per Province (`bysort province: gen prov_sample_propotion = ...`)
|  |  +-- Per Municipality (`bysort municipality_id: gen munic_sample_propotion = ...`)
|  |
|  V
+-----+
| 4. Calculate Sampling Weights (in 'smpL_fixed') |
+-----+
|
|  +-- Absolute/Design Weight (Municipality)
|  | (`gen abs_weight_munic = munic_pop / munic_sample_size`)
|  | (Uses fixed sample size per munic.)
|  |
|  +-- Proportional/Balancing Weight (Municipality)
|  | (`gen prop_weight_munic = munic_pop_propotion / munic_sample_propotion`)
|  | (Uses proportions based on fixed sample size)
|  |
|  V
+-----+
| 5. Survey Data Analysis (Using Absolute Weights) |
+-----+
|
|  +-- Estimate Total Pop from Sample
|  | (`egen total_pop = total(abs_weight_munic)`)
|  |
|  +-- Declare Survey Design
|  | (`svyset [pweight = abs_weight_munic]`)
|  |
|  +-- Compare Population ('census') vs. Raw Sample ('smpL_fixed') vs. Weighted Sample ('svy:'):
|  |
|  |  +-- Mean Income (`frame census: mean income` vs `mean income` vs `svy: mean income`)
|  |  +-- Proportion Urban (`frame census: proportion urban` vs `proportion urban` vs `svy:
proportion urban`)
|  |
|  V
+-----+
| 6. Alternative Weighting Analysis (Using Proportional Wt) |
+-----+
|
|  +-- Compare Total Income:
|  |
|  |  +-- Population (`frame census: sum income`)
|  |  +-- Raw Sample (`sum income` in 'smpL_fixed')
|  |  +-- Prop. Weighted Sample (`sum income [aweight=prop_weight_munic]` in 'smpL_fixed')

```

3. Multistage Sampling (SRS Municipality -> SRS Individual)

```

*****
* Multistage sampling
*****

frame change census

```

```

capture {
    * Switch back to the default frame
    frame change default
    * Drop the 'smpl20' frame if it exists
    frame drop smpl_mult_stg
}
frame put *, into(smpl_mult_stg)

capture {
    * Switch back to the default frame
    frame change default
    * Drop the 'smpl20' frame if it exists
    frame drop temp
}
frame put *, into(temp)

*Randomly selection 300 municipalities
frame change temp
collapse (count) province, by(municipality_id)
drop province
sample 300, count
gen smpld = 1

*merging and keeping randomly selected municipalities
frame change smpl_mult_stg

frlink m:1 municipality_id, frame(temp)
frget smpld, from(temp)
keep if smpld == 1
drop temp smpld

bysort municipality_id: sample 200, count

gen sample_size = _N

bysort province: gen prov_sample_size = _N
bysort municipality_id: gen munic_sample_size = _N

bysort province: gen prov_sample_propotion = prov_sample_size / sample_size
bysort municipality_id: gen munic_sample_propotion = munic_sample_size / sample_size

gen abs_weight_munic = (753/300) * (munic_pop/munic_sample_size)
gen prop_weight_munic = (753/300) * (munic_pop_propotion/munic_sample_propotion)

* --- Survey Data Analysis ---
egen total_pop = total(abs_weight_munic)

svyset [pweight = abs_weight_munic]
frame census: mean income
mean income
svy: mean income

```

```
. frame census: mean income
```

```
Mean estimation                               Number of obs = 1,000,000
```

	Mean	Std. err.	[95% conf. interval]	
income	391705.3	269.7105	391176.7	392233.9

```
. mean income
```

```
Mean estimation                               Number of obs = 60,000
```

	Mean	Std. err.	[95% conf. interval]	
income	360013.2	1038.125	357978.4	362047.9

```
. svy: mean income
```

```
(running mean on estimation sample)
```

```
Survey: Mean estimation
```

```
Number of strata =      1          Number of obs   =    60,000
Number of PSUs   = 60,000          Population size = 1,007,597
                                   Design df       =    59,999
```

	Mean	Linearized std. err.	[95% conf. interval]	
income	394974.9	1209.508	392604.2	397345.5

```
frame census: proportion urban
```

```
proportion urban
```

```
svy: proportion urban
```

```
* --- using weight in regression ---
```

```
frame census: reg income i.urban education
```

```
reg income i.urban education
```

```
svy: reg income i.urban education
```


. frame census: reg income i.urban education

Source	SS	df	MS	Number of obs	=	1,000,000
Model	1.0931e+16	2	5.4653e+15	F(2, 999997)	=	88416.03
Residual	6.1813e+16	999,997	6.1813e+10	Prob > F	=	0.0000
				R-squared	=	0.1503
				Adj R-squared	=	0.1503
Total	7.2744e+16	999,999	7.2744e+10	Root MSE	=	2.5e+05

income	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
urban						
Urban	172412.4	511.3628	337.16	0.000	171410.1	173414.6
education	50499.03	247.603	203.95	0.000	50013.74	50984.33
_cons	252591	449.2588	562.24	0.000	251710.5	253471.5

. reg income i.urban education

Source	SS	df	MS	Number of obs	=	60,000
Model	5.4708e+14	2	2.7354e+14	F(2, 59997)	=	4924.55
Residual	3.3326e+15	59,997	5.5546e+10	Prob > F	=	0.0000
				R-squared	=	0.1410
				Adj R-squared	=	0.1410
Total	3.8797e+15	59,999	6.4662e+10	Root MSE	=	2.4e+05

income	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
urban						
Urban	160350.6	2009.17	79.81	0.000	156412.6	164288.6
education	46731.37	959.8881	48.68	0.000	44849.98	48612.75
_cons	238571.7	1696.017	140.67	0.000	235247.5	241895.9

```
. svy: reg income i.urban education
(running regress on estimation sample)
```

Survey: Linear regression

Number of strata = 1
Number of PSUs = 60,000

Number of obs = 60,000
Population size = 1,007,597
Design df = 59,999
F(2, 59998) = 4084.84
Prob > F = 0.0000
R-squared = 0.1468

income	Linearized					
	Coefficient	std. err.	t	P> t	[95% conf. interval]	
urban						
Urban	168731.1	2401.665	70.26	0.000	164023.9	173438.4
education	50551.82	1122.637	45.03	0.000	48351.45	52752.2
_cons	255826.7	1738.713	147.14	0.000	252418.8	259234.6

```
+-----+
| [Stata Script: Multistage Sampling] |
+-----+
|
| V
+-----+
| 1. Frame Preparation |
+-----+
|
| +-- Clean up old 'smpl_mult_stg' frame
| +-- Create 'smpl_mult_stg' frame from 'census' (will hold final sample)
| +-- Clean up old 'temp' frame
| +-- Create 'temp' frame from 'census' (for PSU selection)
|
| V
+-----+
| 2. Stage 1: Select Municipalities (PSUs) |
+-----+
|
| +-- Switch to 'temp' frame
| +-- Create dataset of unique municipalities (`collapse ... by(municipality_id)`)
| +-- Sample 300 municipalities (`sample 300, count`)
| +-- Mark selected municipalities (`gen smpld = 1`)
|
| V
+-----+
| 3. Merge & Filter Data by Selected PSUs |
+-----+
|
| +-- Switch to 'smpl_mult_stg' frame
| +-- Link to 'temp' frame (`frlink`)
| +-- Get selection marker (`frget smpld`)
| +-- Keep only observations in selected municipalities (`keep if smpld == 1`)
|
```

```

+-- Clean up (`drop temp smpld`)
|
V
+-----+
| 4. Stage 2: Select Individuals (SSUs) |
+-----+
|
|-- Within each selected municipality:
|   `bysort municipality_id: sample 200, count`
|   (Sample 200 individuals)
|
V
+-----+
| 5. Sample Analysis (in 'smpl_mult_stg') |
+-----+
|
|-- Calculate Sample Sizes (Total, Province, Municipality)
|   (`gen sample_size`, `bysort ... gen ..._size`)
|
|-- Calculate Sample Proportions (Province, Municipality)
|   (`bysort ... gen ..._propotion`)
|
V
+-----+
| 6. Weight Calculation (in 'smpl_mult_stg') |
+-----+
|
|-- Absolute Weight (Reflects both stages)
|   `gen abs_weight_munic = (753/300) * (munic_pop/munic_sample_size)`
|   (Inverse probability: PSU selection * SSU selection)
|
|-- Proportional Weight (Adjusted for proportions)
|   `gen prop_weight_munic = (753/300) * (munic_pop_propotion/munic_sample_propotion)`
|
V
+-----+
| 7. Survey Data Analysis (Means/Proportions) |
+-----+
|
|-- Estimate Total Pop (`egen total_pop`)
|
|-- Declare Survey Design (`svyset [pweight = abs_weight_munic]`)
|
|-- Compare Population vs. Raw Sample vs. Weighted Sample:
|   |-- Mean Income
|   |-- Proportion Urban
|
V
+-----+
| 8. Regression Analysis |
+-----+
|
|-- Compare Regression (income ~ urban + education):
|   |-- Population (`frame census: reg ...`)
|   |-- Raw Sample (`reg ...`)
|   |-- Weighted Sample (`svy: reg ...`)

```

Absolute/Design Weight

- **Definition:** This weight is used to correct for unequal probabilities of selection in a survey sample.
- **Formula:**
$$\text{Design Weight} = \frac{1}{\text{Probability of Selection}} = \frac{\text{Population Size}}{\text{Sample Size}}$$
- **Purpose:** Ensures that each unit in the sample represents the correct number of units in the population, compensating for any over- or under-sampling.
- **When to Use:** Use design weights when you have intentionally oversampled or undersampled certain groups within your population. For example, if you oversample

minority groups to ensure adequate representation, design weights will adjust for this oversampling[1].

Proportional/Balancing Weight

- **Definition:** This weight adjusts the sample to match the population proportions.
- **Formula:** $\text{Proportional Weight} = \frac{\text{Population Proportion}}{\text{Sample Proportion}}$
- **Purpose:** Balances the sample to reflect the true distribution of the population, correcting for any discrepancies between the sample and the population.
- **When to Use:** Use proportional weights when your sample does not accurately reflect the population's demographic proportions. For instance, if your sample has more females than males compared to the population, proportional weights will correct this imbalance[2].

Absolute/Design Weight Example

Suppose we have a population of 10,000 people and we select a sample of 500 people.

- **Population Size (N):** 10,000
- **Sample Size (n):** 500
- **Probability of Selection:** $\frac{n}{N} = \frac{500}{10,000} = 0.05$
- **Design Weight:** $\frac{1}{\text{Probability of Selection}} = \frac{1}{0.05} = 20$

Each person in the sample represents 20 people in the population.

Proportional/Balancing Weight Example

Suppose the population is 60% female and 40% male, but our sample is 50% female and 50% male.

- **Population Proportion (Female):** 0.60
- **Sample Proportion (Female):** 0.50
- **Proportional Weight (Female):** $\frac{\text{Population Proportion (Female)}}{\text{Sample Proportion (Female)}} = \frac{0.60}{0.50} = 1.2$
- **Population Proportion (Male):** 0.40
- **Sample Proportion (Male):** 0.50
- **Proportional Weight (Male):** $\frac{\text{Population Proportion (Male)}}{\text{Sample Proportion (Male)}} = \frac{0.40}{0.50} = 0.8$

In this case, females in the sample are weighted by 1.2 to reflect their higher proportion in the population, while males are weighted by 0.8 to reflect their lower proportion.

Multistage Sampling Weight Formula

1. **First Stage Weight (e.g., selecting clusters like schools):** $W1 = \frac{1}{P1}$ where ($P1$) is the probability of selecting the cluster.
2. **Second Stage Weight (e.g., selecting units within clusters like classrooms within schools):** $W2 = \frac{1}{P2}$ where ($P2$) is the probability of selecting the unit within the cluster.
3. **Overall Weight:** $W = W1 \times W2$

Example

Suppose we have a two-stage sampling process:

1. **First Stage:** Select schools with a probability proportional to size (PPS).
 2. **Second Stage:** Select students within each selected school with equal probability.
- **First Stage:** Probability of selecting a school (PPS): $P_1 = \frac{\text{Number of selected schools}}{\text{Total number of schools}}$
 - **Second Stage:** Probability of selecting a student within a school: $P_2 = \frac{\text{Number of selected students}}{\text{Total number of students in the selected school}}$
 - **First Stage Weight:** $W1 = \frac{1}{P1}$
 - **Second Stage Weight:** $W2 = \frac{1}{P2}$
 - **Overall Weight:** $W = W1 \times W2$

Numerical Example

1. **First Stage:** Select 10 schools out of 100. $P1 = \frac{10}{100} = 0.1$ $W1 = \frac{1}{0.1} = 10$
2. **Second Stage:** Select 20 students out of 200 in each selected school. $P2 = \frac{20}{200} = 0.1$ $W2 = \frac{1}{0.1} = 10$
3. **Overall Weight:** $W = W1 \times W2 = 10 \times 10 = 100$

Each selected student represents 100 students in the population.